

## **Experiment 8: Practical Exercise: Set Up a Jenkins CI Pipeline for a Maven Project, Use Ansible to Deploy Artifacts Generated by Jenkins.**

**Step 1: verify java, maven, jenkins and ansible are installed**

**Step 2: Choose maven project and execute the following on terminal**

```
sudo chown -R jenkins:jenkins /home/klecet/2kd23cs094
```

```
sudo chmod -R 755 /home/klecet/2kd23cs094
```

**Step 3: Login to the jenkins**

Go to manage Jenkins-plugins-available plugins-maven integration-pipeline-junit-ansible

**Step 4:Go to manage Jenkins-tools-maven installation-add maven installation-uncheck install automatically-Enter**

Name

**Apache Maven 3.8.7**

MAVEN\_HOME

**/usr/share/maven**

**Click on Save**

**Step 5: Create Jenkins Job**

1.Go to Jenkins Dashboard

2.Click **New Item** (Give some name)

3.Select **Freestyle Project**

4.Click **OK**

**Build Section:**

Click **Add Build Step** → **Invoke top-level Maven targets**

Enter: **clean package**

Advanced: **/home/klecet/2kd23cs094/pom.xml**

**Save and build now.**

**Step 6:Run on Terminal**

**nano inventory.ini**

Add the following:

**[local]**

**localhost ansible\_connection=local**

### Step 7: Run on Terminal

**nano deploy.yml**

Add the following:

```
---  
- name: Deploy Maven Artifact  
  hosts: local  
  become: yes  
  
tasks:  
  - name: Create deployment directory  
    file:  
      path: /opt/deployment  
      state: directory  
      mode: '0755'  
  
  - name: Copy artifact from Jenkins workspace  
    copy:  
      src: /home/klecet/2kd23cs094/target/2kd23cs094-1.0-SNAPSHOT.jar  
      dest: /opt/deployment/  
      mode: '0755'  
      remote_src: yes
```

### Step 8: Run on Terminal

**ansible-playbook -i inventory.ini deploy.yml**

### Step 9: Integrate Ansible with Jenkins

Go to:

Your item name → Configure

**Build Steps:**

Click:

**Add Build Step → Execute Shell**

Add:

**ansible-playbook -i /home/klecet/inventory.ini /home/klecet/deploy.yml**

Click **Save and build now**.

### Step 10: Run on terminal to find IP address of your system

**hostname -I**

### Step 11: Run on terminal

```
sudo nano /etc/nginx/sites-available/jenkins
```

Add:

```
server {  
  
    listen 80;  
  
    server_name localhost 127.0.0.1 172.16.6.38 ;  
  
    location / {  
  
        proxy_pass http://localhost:8080;  
  
        proxy_set_header Host $host;  
  
        proxy_set_header X-Real-IP $remote_addr;  
  
        proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;  
  
        proxy_set_header X-Forwarded-Proto $scheme;  
  
    }  
  
}
```

### Step 12: Run on terminal

```
sudo ln -s /etc/nginx/sites-available/jenkins /etc/nginx/sites-enabled/
```

### Step 13: Run on Terminal

```
sudo nginx -t
```

### Step 14: Run on Terminal

```
systemctl status nginx
```

### Step 15: Go to Browser and type

<http://localhost> (nginx redirects to jenkins page)